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Differential Adsorption and Chemical Affinities of Proteins for Apatitic Surfaces

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Studies are reviewed which identify the interacting groups involved in protein adsorption onto hydroxyapatite. Thus, carboxyl and phosphoester groups bind to calcium sites in the adsorbent, and basic groups bind to phosphate sites. Detailed adsorption studies have been performed to quantitate some of these interactions. An adsorption model, based on the Langmuir adsorption isotherm, adequately fitted the data from experiments using selected amino acids, bovine albumin and two human salivary proteins. Adsorption parameters (affinities and maximum number of sites) were obtained for several apatitic adsorbents, with affinities increasing considerably in the series hydroxy- (HA), fluorhydroxy- (FHA) and fluorapatite (FA). A modest increase in the number of sites was also noted. The change in adsorption behavior, with increasing fluoride content, was attributed to a reduction in the surface free energy of the adsorbent, with a concomitant decrease in the interaction of the adsorbent with water, and a consequent enhancement of the adsorption bond. It is suggested that this effect may play a role in the cariostatic effect of fluoride. Unusual structural aspects of the salivary proteins are discussed in relation to their adsorption behavior, and the molecular segments responsible for binding to the adsorbent tentatively identified.

Introduction.

An intimate association of mineral and organic macromolecular constituents is a characteristic feature of mineralized tissues. The macromolecular components play important modifying roles, both during mineralization and in determining the final properties of the tissue. The molecular basis of these macromolecule/mineral interactions, however, is not clearly understood. Similarly, although chromatography on HA columns has been extensively used to purify macromolecules[1], the basis for the separations obtained is unclear, and the chromatographic behavior of proteins correlates poorly with their molecular properties[2]. Studies of a phosphoprotein from dentin[3] have shown it to affect the crystallinity of calcium phosphate precipitates[4], and phosphoproteins from saliva have been shown to inhibit spontaneous and seeded precipitation from solutions supersaturated with respect to calcium phosphate salts[5,6], though the mechanisms by which they act are not yet understood. These latter studies have a particular bearing on the interactions which occur between dental enamel and salivary proteins, to form a protein integument of the tooth surface[7,8], but the detailed mechanisms by which this structure forms are unknown. Some steps, however, have been taken towards identifying the nature of the reactive groups involved in these interactions[9,10,11], and recently the first actual measurements of adsorption affinities of selected amino acids[12] and macromolecules[13,14] for HA were reported. The purpose of this paper is to bring together results from studies dealing with interrelationships between macromolecules and calcium phosphate salts, and to provide a review of present knowledge concerning protein-HA interactions relevant to dental science.

Nature of the adsorption centers of proteins and HA.

Early studies failed to yield correlations between the chromatographic behavior of proteins on HA columns, and their molecular properties such as net charge, proportions of acidic and basic amino acids, isoelectric points or molecular size[2]. A single exception was the finding[9] that the presence of ester phosphate groups resulted in exceptionally strong binding of proteins to HA during chromatography, buffers of relatively high molarity being required for the elution of phosphoproteins. A later systematic study[10] of synthetic polymers of amino acids showed the importance of free carboxyl groups. Thus, homo- and heteropolymers containing aspartate and glutamate residues required buffer concentrations for elution in approximate proportion to their dicarboxylic amino acid content, whereas polymers lacking these residues, or where the carboxyl groups

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were esterified, were not retarded during chromatography. It was concluded that the mechanism of carboxyl and phosphoester binding to HA was by a direct interaction between calcium ions on the HA surface and the anionic groups in the macromolecule[10]. It was also argued that elution occurred as a consequence of displacement of the binding anionic group from the adsorption site, by the eluting buffer anions, and that ionic strength effects played minor roles. Thus, the elution of polyglutamate, by means of a phosphate gradient, occurred at the same phosphate concentration both in the presence, and absence, of 1M KCl.

A subsequent report[11] showed that basic residues also interact with HA, thus correcting a previous negative conclusion[10]. In fact, poly(L-lysine), poly(L-arginine) and poly(L-ornithine), having molecular weights of 7,000 or greater, could not be eluted using 1M potassium phosphate, but were eluted by higher concentrations of monovalent electrolytes, with the cation being the active eluting species and the anions having little effect. Since basic polymers were relatively easily eluted by low concentrations of divalent cations, calcium and magnesium being almost equally effective, it was concluded that basic residues interacted with phosphate sites on the HA surface, from which they could be competitively displaced by high concentrations of monovalent cations, or, more effectively, by much lower concentrations of divalent cations; hence, adsorption is electrostatic in nature.

The behavior of native proteins[10,11] appears in most cases to be qualitatively understandable in terms of the foregoing generalizations. Accurate quantitative predictions, however, appear to be difficult to make, and many exceptions and anomalies occur. This seems consistent with the likelihood that behavior of the macromolecules will be strongly influenced by their secondary and tertiary structures. Thus, polyglutamate showed the same chromatographic behavior in the absence and presence of 8M urea. Native proteins, on the other hand, behaved quite differently after denaturation[11]. Lysozyme, ribonuclease and cytochrome C in 8M urea were either not retarded on HA columns, or were eluted more easily than the native proteins. Tropocollagen, denatured by heating, gave products in which the proportion of non-retarded material increased with increasing temperature, and in which the retarded constituents required buffers of progressively lower molarities for elution. These results were explained in terms of clusters of charged residues on the surface of the native molecule, thus increasing the probability of binding, with this conformation being lost on denaturation.

Further evidence supporting the concept of interactions between phosphate and carboxylate ions and calcium ion sites, and interactions between amino and guanidinium groups and phosphate sites in HA, has come from electron spin resonance studies of calcium phosphate-macromolecule co-precipitates[15,16]. Using vanadyl ions as reporter groups for calcium ion interactions, and x-irradiation at low temperatures, followed by measurement of spin energy transfer between macromolecule and mineral led to confirmation of carboxylate ion and phosphoester-calcium ion interactions, and guanidinium and amino group-phosphate ion interactions, among others.

Adsorption model: studies of adsorption of selected amino acids.

The foregoing studies yielded important qualitative information about the nature of the interacting groups, but the strength of an interaction could only be described quantitatively in arbitrary terms, such as the buffer concentration required for the elution of a particular macromolecule. While this enabled distinctions to be made between the strong interactions occurring between phosphoester groups and HA, and the relatively weaker interactions of carboxyl and basic groups, classical techniques, *e.g.*, adsorption isotherms, for quantitatively describing such interactions were only recently applied to this problem. Interpretation of macromolecule adsorption isotherms and the elucidation of the adsorption mechanisms in physical/chemical terms are, however, exceptionally complex problems. Since several different functional groups of the macromolecule may be involved, and at least two possible adsorption sites exist on the apatite surface, simple compounds were chosen for initial studies[12], in which only a limited number of functional groups could be involved. Adsorption isotherms were obtained for aspartic, glutamic, DL-

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α-aminoadipic, DL-α-aminopimelic, succinic and adipic acids and arginine, by measurement of the amount adsorbed per gram of hydroxyapatite, Q (μM/g), at various equilibrium concentrations, C (μM/ml). The results were compatible with an adsorption model which closely matched the Langmuir adsorption isotherm[17,18], *i.e.*,

$$\frac{C}{Q} = \frac{1}{KN} + \frac{C}{N} \quad \bullet \bullet \bullet \bullet \qquad [1]$$

where N is the number of adsorption sites per gram of adsorbent, and K is a parameter, usually termed the affinity constant, which reflects the strength of the bond between the adsorbate molecules and the adsorption sites. In cases where comparison of different adsorbents is required, Q is expressed in terms of adsorbed solute per unit surface area.

In agreement with this model, plots of C/Q against C gave straight lines, shown in Figure 1. From the slopes and intercepts of these were calculated values for the number of adsorption sites, N, and the affinity constants, K. These adsorption parameters, listed in Table 1, lead to several interesting conclusions. Thus, the adsorption affinities of the α-amino acids are very similar, indicating that the same functional groups in the amino acid, and similar sites in the adsorbent, are involved in the adsorption phenomenon. It is reasonable to identify the terminal carboxyls of the adsorbates, and calcium ions in the adsorbent surface, as the reactive entities. The number of sites decreases, in a systematic way, with increasing molecular size. Based on the reported geometry[20] of aspartic and glutamic acid molecules, and taking into account the crystallographic characteristics and geometry of the HA crystals, it was possible to develop a geometric model for the adsorption of the acids tested. A model in which the long axes of the molecule are arranged perpendicularly or horizontally to the surface was not compatible with the experimental data. Rather, the data matched a model where the molecules are anchored by their terminal carboxyl groups, and are inclined at an angle to the surface. Rotation of the molecule generates a cone, and its projection onto the surface defines the extent of the adsorption site. An angle of 55° was adopted for aspartic acid, and 40° for the other molecules, and the resulting model geometry and molecular dimensions were used to calculate the number of sites available for each of the acids. The results of this calculation, shown in Table I, gave a rather satisfactory matching considering uncertainties inherent in the assumptions which were made. In spite of this agreement, caution is still required in accepting its validity. Small, but significant deviations from predicted behavior were noted. Thus, at high adsorbate concentrations, the isotherms of aspartic acid and glutamic acid deviated slightly from predicted behavior. This anomaly was explained in terms of a small degree of inhomogeneity of the adsorbent surface, with the possible consequence that all adsorption sites were not isoenergetic, so violating a requirement for Eq.[1]. However, if the results for the amount adsorbed at the low concentration range of the isotherm (Q), are normalized with respect to the number of adsorption sites (N), and the resulting values of Q/N (*i.e.* the fractional number of sites occupied by adsorbate molecules) plotted against the corresponding equilibrium concentrations, all the points describe the same function, with the possible exception of aminopimelic acid[12]. This indicates that the adsorption of all the molecules involves the same energy levels, and that the adsorption differences between the individual members of the series are simply due to differences in availability of adsorption sites, which appear to be dictated by geometric considerations.

A further anomaly was that coverage of the adsorbent surface, by the adsorbates tested, was low compared to the apparently available surface, determined by nitrogen adsorption. At least two factors may contribute to this discrepancy. First, nitrogen molecules would have easier access to surfaces, *e.g.* interstices and defects from which larger molecules would be excluded. Second, there are probably vacancies at calcium sites. Treatment of the adsorbent with calcium chloride substantially increased adsorption of aspartic acid (Table 1). Similarly, adsorption of arginine is undetectable, unless the adsorbent is first treated with phosphate, suggesting the presence of vacancies (or defects) in the crystal surface. Non-specific enrichment of the surface, however, unrelated to filling vacancies, would also be consistent with the results obtained.

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It seemed significant that the proposed adsorption model was also able to accommodate two dicarboxylic acids (Table 1). The affinity constants are substantially higher, consistent with the higher probability for adsorption since the two carboxyl groups are equivalent, and good agreement was obtained between the experimentally determined number of sites, and the value predicted from the respective models.

Although the foregoing model for amino acid adsorption onto HA is still capable of further refinement, the agreement between the model and experimental data and the predictive value of the geometric model suggests that a relatively good understanding of the system has been obtained. The small deviations from predicted behavior, noted above, do not significantly impair the model; rather they indicate the value of using well-characterized, highly-purified simple compounds as probes to investigate and characterize the HA crystal surface. This kind of preliminary study seemed a necessary step before embarking on similar studies using complex macromolecules, the results of which are not so amenable to the kind of detailed analysis described above.

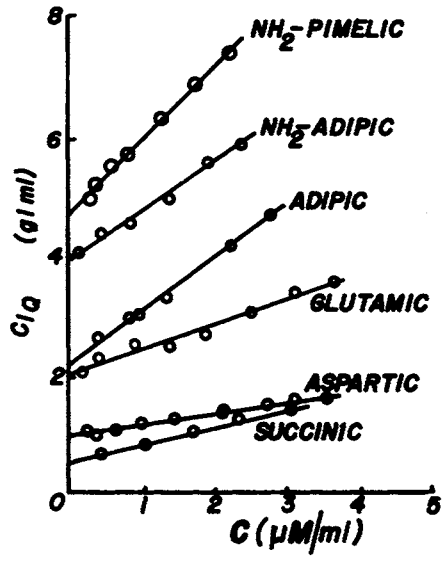


Fig. 1 – Adsorption isotherms plotted according to Eq. [1].

Significance of selected macromolecule/mineral interactions.

Although numerous biological systems exist in which macromolecule/HA interactions occur, few studies have been concerned with the interactions between the individual purified macromolecules, and well-characterized apatitic minerals. One system, for which some relevant studies have been performed, is the human tooth-saliva interface. The enamel surface, when exposed to saliva, acquires a well-defined integument[7,8,20], usually termed the acquired enamel pellicle[21]. Histologically, the 0.1-1.0μm thick pellicle[22] generally lacks structure, though older pellicles may possess a banded appearance. Analytical studies[23,24] suggest that it forms from salivary proteins by a selective adsorption process, this being consistent with the observed selective adsorption of salivary proteins onto HA from *in vitro* systems[25,26,27,28]. The pellicle is biologically significant in that it strongly affects the initial attachment of microorganisms to the tooth surface[29,30,31] and remains interposed between the enamel and dental plaque. It reduces the extent of demineralization of enamel, both by acidic buffers[32,33,34] and cariogenic microorganisms *in vitro*[35]. This may be the basis of its proposed ability to reduce the severity of carious attack[36,37] *in vivo*. Recently it has been suggested that the pellicle may have a further property in that some of its constituents block nucleation sites on the enamel surface, and so prevent the formation of unwanted mineral accretions on the teeth from the saliva, which is normally supersaturated with respect to the enamel mineral[38]. These properties, both biologically interesting, and important for health reasons, have led to systematic studies of the interactions between salivary proteins and apatitic minerals. A particular area of interest is the possibility of different adsorption behavior onto HA, FHA, and FA, differences which may affect the nature of the pellicle and contribute to the caries-reducing properties of fluoride.

Clearly, a 0.1-1.0μm thick pellicle is able to accommodate many layers of protein molecules. Most attention has been focused, however, on the possible specificity of proteins which may form the first monolayer on the apatite surface. This specificity could be determined by selective adsorption onto the mineral surface. Experiments specifically

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TABLE I
ADSORPTION PARAMETERS FOR AMINO- AND DICARBOXYLIC ACIDS

Adsorbate	Affinity, ml/μM K in Eq. [1]	Number of sites, N. μM/g	
		Calculated from Eq. [1]	Geometric model
L-Aspartic acid	0.203	53.5	53.5*
L-Glutamic acid	0.206	24.4	26.6
DL-α-aminoadipic acid	0.218	11.9	13.3
DL-α-aminopimelic acid	0.275	7.85	10.0
Succinic acid	0.558	35.3	35.7
Adipic acid	0.425	11.1	13.3
Pretreated HA			
L-Aspartic acid ($CaCl_2$)	0.164	72.1	
L-Arginine (pH7.8 phosphate buffer)	9.47	0.635	

*Aspartic acid was used as the reference molecule for calculating the remaining values of N for the geometric model.

designed to reveal selective adsorption effects[39,40] have identified several "apatite-reactive" proteins. Thus, by equilibrating different amounts of adsorbent with constant amounts of saliva, it was possible to determine the mineral/saliva ratio at which only a single constituent adsorbed[40]. This was identified as an unusual tyrosine-rich phosphopeptide[41], which has been named statherin, the complete primary structure of which was recently reported[6]. This peptide could be nearly completely adsorbed from glandular salivas before significant adsorption of other constituents occurred. As statherin was depleted from solution, a histidine-rich phosphopeptide was next removed[42], this being followed by the adsorption of a group of four unusual proline-rich phosphoproteins[43,44,45], partial sequences of which have recently been reported[46,47]. In parallel with the adsorption of these acidic macromolecules were adsorbed small amounts of a low molecular weight basic peptide, though whether this was a specific effect has yet to be determined. This combination of well-defined selective adsorption, progress in determining the molecular structure of the adsorbates[6,46,47], and the detailed adsorption studies which have been conducted[13,14], make this system particularly amenable to analysis. An additional aspect of these studies was the discovery that statherin and the proline-rich proteins possessed the unusual ability to modify the solution and solid phase properties of calcium phosphates. Thus the proline-rich proteins and statherin were found to be effective inhibitors of both seeded precipitation[59] and of the reactions leading to the transformation of acidic salts, such as dicalcium phosphate dihydrate, to basic calcium phosphates[48]. In addition, statherin was shown to be a particularly potent inhibitor of precipitation from solutions which are strongly supersaturated with respect to calcium phosphate salts[5,49]. It seems that the inhibitory activities of these proteins, and their selective adsorption behavior, are likely to have similar bases, perhaps in some feature of their molecular structure. Also, since these effects clearly depend on interactions between a variety of calcium phosphate surfaces, and the macromolecules concerned, a systematic study of adsorption of the macromolecules onto well-defined calcium phosphate surfaces would be expected to reveal important details of the interactions involved.

Macromolecule adsorption studies.

Adsorption studies of bovine serum albumin, statherin and one of the proline-rich proteins, PRP-3[43], and a specific molecular segment obtained by tryptic digestion have recently been obtained[57]. Essential prerequisites for these studies are pure, well-characterized adsorbents, purified macromolecules, and measurements made at the equilibrium state. The preparation of pure,

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well-crystallized HA, FA and a series of fluorhydroxyapatites has been reported previously[50]. In the studies to be considered, two FHA preparations (13 and 41% substitution of OH^- by F^-) were used. The adsorbents were all extensively characterized[14]. Under the polarizing microscope, all four apatites appeared mostly as rosettes formed by acicular crystals about 3-6μm long by about 2-3μm thick. Some crystals ranged up to 10-15μm long by 5-7μm thick. The crystals displayed parallel extinction and had a mean index of refraction (white radiation, 25°C) of 1.636± 0.002. The x-ray diffraction patterns for all four samples were sharp and characteristic of calcium apatites and the *a* and *c* crystallographic parameters for HA and FA, determined from the spectra[50], agreed well with the literature values[51,52]. Other analytical data and surface areas have been reported[14].

The adsorbents were all well-purified proteins. These were a crystalline bovine serum albumin, a highly purified preparation of statherin[6], shown to be pure by electrophoretic, immunochemical, amino-terminal and chromatographic analyses, and proline-rich proteins 1 and 3[43] which were similarly characterized except that no immunochemical analyses were performed.

Two steps were taken to ensure that measurements were made under conditions as close to equilibrium as practically possible. First, the solvent (0.05M NaCl) was saturated with respect to adsorbent before addition of the adsorbate. Second, time courses of adsorption were obtained to determine the time required for the adsorption process to reach equilibrium, so determining the experimentally-used equilibrium time. There were two reasons for using 0.05M NaCl as solvent; first, it approximated the ionic strength of saliva, and second, it was required to maintain statherin in solution at the desired concentrations, this macromolecule being less soluble in water than in the dilute electrolyte solution. The experimental system used for the salivary proteins consisted, therefore, of a series of accurately weighed samples of the mineral adsorbates, preequilibrated with solvent (20 hr), to which appropriate samples of adsorbate were added. Further equilibration (4 hr) was performed before the concentration of adsorbate remaining in solution was determined, and the adsorption isotherms obtained, as described above for the amino acids.

Initial adsorption studies, using bovine serum albumin and aspartic acid[13], were conducted for two reasons. First, it was necessary to test the experimental system using a protein, and so determine if the approach, successfully used for amino acids, was suitable for examining macromolecules. The second reason was to conduct experiments using an available well-defined macromolecule and an amino acid, to determine if differences in adsorption behavior existed between HA, FHA and FA for small and more complex molecules. Such preliminary experiments seemed necessary, before embarking on studies requiring large amounts of purified salivary proteins.

Plots of C/Q vs. C for the two adsorbates gave linear relationships[13]. The linearity of the plots extends the validity of this model to the adsorption of aspartic acid onto FHA and FA, and shows its validity for the adsorption of bovine albumin onto all three adsorbents. The greater scatter of the points for the protein presumably reflects the greater analytical difficulties involved in determining protein concentrations. The adsorption parameters, K and N, calculated from the linear plots are given in Table 2. In general terms, the presence of fluoride as a lattice constituent increases both the maximum number of sites, N, and the affinity, K, for the two molecules tested. The observed increase in the value of N may be related to better crystallinity, with increasing fluoride content, possibly resulting in fewer calcium vacancies at the crystal surface, with a concomitant increase in number of sites. The data, however, strongly indicate that the fluoride itself is contributing to the increase in the number of sites, and to the strength of the adsorption bond. Conceivably, the enhancement in adsorption may involve an interaction between fluoride sites and the amino group of aspartic acid, either alone, or in conjunction with a calcium-carboxyl bond. Bovine albumin, as with aspartic acid, shows an increase in the number of sites with increasing fluoridation, and a substantial increase in affinity. There seems, however, to be an upper limit for this increase, at least for bovine albumin. A possible explanation may be that fluoride contributes to albumin attachment by the possible increase in calcium sites, noted

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TABLE 2
ADSORPTION PARAMETERS FOR ASPARTIC ACID AND BOVINE ALBUMIN

Adsorbent	Affinity, K. ml/μM	Max. number of sites, N. μM/m²
Adsorbate: L-Aspartic acid		
HA	0.30	0.44
FHA (41%)	1.50	0.72
FA	3.76	0.68
Adsorbate: bovine albumin		
HA	24,900	0.028
FHA (41%)	58,000	0.033
FA	44,900	0.041

TABLE 3
ADSORPTION PARAMETERS FOR TWO HUMAN SALIVARY MACROMOLECULES

Adsorbate	Adsorbent	Affinity, K. ml/μM×10⁻³	Max. number of sites N. μM/m²
PRP-3	HA	26.0	0.16
	FHA (13%)	42.1	0.16
	FHA (41%)	44.0	0.19
	FA	139	0.20
Statherin	HA	11.1	0.49
	FHA (41%)	20.0	0.52
	FA	33.6	0.57
Statherin (in 10^{-3}M NaF)	HA	124	0.48
	FA	58	0.57

above, and by providing negative fluoride attachment centers for basic groups in the molecule. At high surface fluoride contents, however, this effect may be negated by an increased electrostatic repulsion of the predominately negatively-charged albumin molecule. In any case, this effect has been seen only with bovine albumin, and does not occur with the salivary proteins, discussed below. Thus, specific structural features of the albumin molecule may be responsible for the observed behavior.

Studies using human salivary statherin and proline-rich proteins.

The foregoing studies validated the experimental approach, and the proposed adsorption model revealed significant differences between the properties of the adsorbents, and led to a detailed study of the adsorption behavior of two salivary proteins on the same adsorbents[14]. As with the previous studies, plots of C/Q vs. C for statherin and PRP-3 were linear, so confirming the validity of the model for these proteins. Numerical values of K and N, calculated from the slopes and intercepts of the isotherms, are given in Table 3. The affinities of statherin for the adsorbents tested are less than one-half the corresponding values for PRP-3. For both proteins, the affinities increase markedly with an increasing fluoride content in the lattice, with the five-fold increase in affinity for PRP-3 being particularly striking. The maximum number of adsorption sites, N, for statherin is more than double the corresponding value for PRP-3. Only a modest increase in the number of sites was obtained for both adsorbates, with increasing fluoride content. It is also noteworthy that quite low levels of fluoride in the lattice (13% F^- sites, equivalent to 4,400 ppm F^-) markedly affect the extent of site occupancy.

It seems very significant that all adsorbents tested show a greater affinity for fluoride-containing apatites than for HA.

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This seems to be a generalized phenomenon for small (aspartate) and large (albumin) molecules. A probable explanation for this behavior is that the experimental isotherms reported here are composite isotherms[53,54,55], *i.e.* they represent the net result of the competitive adsorption of the solute and its solvent onto the same surface. In all probability, both adsorbate and solvent molecules are adsorbed at equilibrium, and the presence of the solvent molecules weakens the bond, presumably electrostatic, between the adsorbate molecules and the adsorption sites on the surface of the adsorbent. Incorporation of fluoride into the apatite lattice, or into the surface of HA adsorbents, results in a solid with a lower surface energy and, therefore, the interaction with water should be weaker, as indeed has been found experimentally[56]. Thus the interaction between adsorbate and the adsorption site will be augmented.

This concept was tested further by determining the adsorption isotherms of statherin on HA and FA in the presence of 10^{-3}M NaF. The adsorption parameters are shown in Table 3. Clearly the presence of fluoride dramatically increases the affinity of statherin for HA, without affecting the number of sites. An increase was also noted for FA, though this was relatively smaller. The low activities of Ca^{++} and F^- in solution precluded the formation of CaF_2; thus, it must be concluded that the adsorbent surfaces were substantially changed without affecting the basic apatitic structure. The most plausible explanation is that F^- exchanges for OH^-, decreasing the affinity of the adsorbent for water, and so increasing the strength of the adsorption bond. Undoubtedly, other factors are operating as well, otherwise the affinity of statherin for HA, in the presence of 10^{-3}M NaF, should not be greater than its affinity for FA. The observed enhancement, however, may be explained, at least partly, by the presence of defects or vacancies which differ in the two adsorbents. The results also suggest that the FA, suspended in aqueous solutions, may contain fluoride positions occupied by hydroxyls which are displaced by fluoride ions from the solution, so enhancing the bond strength.

The adsorption experiments described above employed well-crystallized and thoroughly characterized adsorbents, highly purified adsorbates and an experimental system in which measurements were made as close to the equilibrium state as practically possible. The results clearly show that the presence of fluoride in the apatite lattice leads to a substantial increase in the strength of the adsorption bond, and to a small, but distinct, increase in the number of adsorption sites. These conclusions may appear to contradict those obtained in previous studies[60,61,62,63] where apatitic adsorbents, treated with fluoride, appeared to possess a lowered capacity for adsorbing various proteins. Some of the reasons underlying these apparent disagreements have already been critically examined[13,64]. Generally, the adsorbents have not been well-defined, either in terms of surface area, or quality of their apatitic structure (stoichiometries resulting in Ca/P=1.5, rather than 1.67). In these studies, it was not intended that measurements would be made close to, or even approaching, equilibrium conditions. One other reason for disagreement centers on the possible modification of the surface of the adsorbents. In studies reviewed here[13,14], the adsorption of albumin and statherin onto HA and FA was examined in the presence and absence of 10^{-3}M NaF. The ionic activity product for CaF_2 in the experimental system was calculated to be 1.85×10^{-11}, compared to the solubility product of CaF_2, 3.58×10^{-11}. Thus, the system was half saturated with respect to CaF_2, so precluding the formation of this salt on the adsorbent surface. A contrary situation exists in a study where HA was treated with the same concentration of NaF, but at a lower ionic strength[63]. The reduction of the modifying effect of the electrolyte on ionic activities would lead to the solubility product of CaF_2 being exceeded, with the virtually inevitable formation of this salt on the adsorbent surface, and a consequent change in adsorption properties.

A further adsorption parameter is the actual surface coverage (as opposed to site occupancy) of the adsorbent surface. Calculation of molecular dimensions of adsorbed molecules, however, involves assumptions regarding both the shape and the orientation of non-spherical adsorbed molecules. Although reasonable coverages were calculated, and explanations developed for the deviations observed, definitive conclu-

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sions regarding coverages require more information regarding molecular structure and shape. This problem will be considered further below.

Adsorption and molecular structure.

It should be possible, ultimately, to define the reactive groups of molecular segments which mediate attachment of macromolecules to the apatitic adsorbents. A prerequisite here is knowledge of the primary structure of the molecule, with an understanding of the secondary and tertiary structures also being important in most cases. The complete primary structure of statherin is known[6], the structures of PRP-1 and 3 nearly completed[46,47] and the structures of important segments of PRP-2 and 4 determined[47]. Studies aimed at identification of the binding sites in statherin and the proline-rich proteins have only recently begun. Thus, removal of phosphate from PRP-1 decreased its ability to bind to HA[46]. Also, it was possible to adsorb a reactive fragment from a tryptic digest of PRP-1 onto HA, the amino acid composition of which corresponded to the amino-terminal fragment[46]. A more specific study[57] led to the measurement of the adsorption affinity of this fragment, this being described below. More information about possible binding sites has come from studies of these proteins, and specific fragments obtained by enzymatic cleavage, with respect to their ability to inhibit precipitation from solutions supersaturated with respect to calcium phosphate salts. It seems almost certain that the molecular segments responsible for this inhibitory activity, and for binding to apatitic surfaces, will be the same.

Statherin has an unusual structure, the first 20 residues of this 43 residue peptide being:

```
    1                   5
+ASP-PSE-PSE-GLU-GLU-LYS-PHE-LEU-
          10                  15
ARG-ARG-ILE-GLY-ARG-PHE-GLY-TYR-
                    20
GLY-TYR-GLY-PRO-
```

Ten of the twelve charge centers present occur in the first 13 residues, the remaining 30 residues being essentially uncharged. A series of fragments obtained by enzymatic cleavage (1-18, clostridiopeptidase A; 1-9, clostridiopeptidase B; 1-7, chymotrypsin; 1-6, trypsin) possessed similar inhibitory activities in a standard assay system[58], suggesting an important role for the segment ASP-PSE-PSE-GLU-GLU-LYS. Removal of phosphate from the two phosphoserines (residues 2 and 3), using an alkaline phosphatase, abolished the activity, indicating a critical role for these vicinal phosphoserine residues. A similar result was obtained from studies of the proline-rich proteins. Amino-terminal tryptic fragments obtained from PRP-1 and 3 have the following identical structures, while the corresponding fragments from

```
1                   5
PCA-ASP-LEU-ASN-GLU-ASP-VAL-PSE-
         10                  15
GLN-GLU-ASP-VAL-PRO-LEU-VAL-ILE-
                   20
SER-ASP-GLY-GLY-PSE-ASP-GLU-GLN-
         25                  30
PHE-ILE-ASP-GLU-GLU-ARG-
```

PRP-2 and 4, which are genetically distinct proteins from PRP-1 and 3, differ in small ways, residue 4, for instance, being ASP in PRP-2 and 4[47]. As with statherin, the amino-terminal tryptic fragment contains both of the phosphoserine residues, and almost all of the anionic residues present in these proteins. This fragment is also the only one to possess inhibitory activity, from which, as with statherin, activity is lost on removal of phosphate from the phosphoserines. Unlike statherin, the phosphoserines, at residues 8 and 21, are well separated, suggesting that they may act independently, but cleavage of the active fragment with thermolysin (presumed to yield the separate phosphopeptides, residues 7-11 and 16-25) diminished inhibitory activity to a very low level, suggesting possible cooperation between these residues in the parent molecule. Measurement of the adsorption affinities of PRP-1, its active amino-terminal tryptic fragment and PRP-3 gave values (24,800, 25,300 and 26,000 respectively) which were the same within experimental error. All these results, taken together, essentially confirm that the segment responsible for

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mediating adsorption, and for inhibitory activity, lies within the first 30 residues of these molecules.

The studies reviewed here were selected for their contribution to an understanding of the interactions occurring between macromolecules and apatitic adsorbents. Information relevant to this problem has come from chromatographic[9,10,11] and physical chemical studies[12,13,14]. The latter investigations were aimed at gaining an understanding of a specific system, *i.e.*, the first steps in the formation of the acquired enamel pellicle, and the investigation of the effect of fluoride on the processes involved. The marked effect of fluoride on adsorption has interesting implications, since the pellicle is known to confer protection against demineralization[35], presumably because of its permselective properties[33]. Pellicles formed on fluoride-rich tooth surfaces may well differ in these properties, with these differences contributing to the cariostatic effect of fluoride. The methods and results of the reviewed studies also have relevance to the interactions occurring in mineralized tissues, both during and after mineralization. Thus, the properties and effects of the dentin phosphoproteins[3,4] have clear parallels with the effects caused by the salivary phosphoproteins[5,6]. In spite of these similarities, caution should be exercised in attempts to interpret the complex macromolecule/mineral relationships of mineralized tissues, formed under cellular control, in the simple physical chemical terms described above.

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Session VII Discussion

Dr. Arends: Could you tell a little bit about the substrates you used, I mean the BET values, particle size and the pretreatment you gave before you started the adsorption procedure?

Dr. Hay: The adsorbents were prepared by Dr. Moreno, who is better able to answer your question.

Dr. Moreno: These apatites were prepared by a very slow precipitation process that takes about eight days, at boiling temperature and high pH in order to minimize formation of more acid phosphates. The optical properties correspond to those of homogeneous apatites. The x-ray diffraction patterns do not show any extraneous materials, and the unit cell parameters calculated from the spectra of hydroxy- and fluorapatite check with the values reported by other investigators. The preparations show a distribution in the crystal size with a maximum length of about 15 um and 2-3 um in width down to lengths of 3-5 um and widths of 1-2 um.

Dr. Arends: And the BET value?

Dr. Moreno: The BET value changes with the fluoride content of the apatite. For pure hydroxyapatite, the specific surface area is in the order of 12 m^2/g whereas for fluorapatite, it is in the order of $6m^2/g$.

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Dr. Arends: Do you have any indication of the presence of acid phosphate groups, on the surface or in the material?

Dr. Moreno: No, we don't.

Dr. Nancollas: It always amazes me that the Langmuir isotherm can be applied to so many systems. We began using it about 1953 for the adsorption of molecules on crystals, which, of course, have a very large number of imperfections. You would not expect an isotherm, which assumes homogeneity of sites to work. In the adsorption of the carboxylates on apatite, I could not see whether the order follows the relative solubilities of the calcium salts or the dissociation constants of the calcium ion pairs.

Dr. Hay: We have not attempted to establish if adsorption behavior correlates with other molecular parameters. All we can say at the moment is that the affinity constants vary only slightly within a series, and this variation may possibly correlate with other variables.

Dr. Gedalia: Both Dr. Hay's and Dr. Moreno's studies have far-reaching clinical implications. I would like to ask a question regarding topical application of fluoride. We are always instructed to remove the pellicle or to remove some microns of the surface enamel before fluoride application is carried out. But, I don't know if this is the right thing to do, if the pellicle protects the enamel. I know that some scientists have recommended that it should not be removed prior to application of a topical fluoride. I would like to know your opinion about it.

Dr. Hay: I think that the important point here is that the pellicle does re-form quite quickly on enamel surface after abrasive or acid treatments. For instance, a protein film, with the characteristic amino-acid composition of pellicle, is formed within two hours, *in vivo*, after abrasive treatment. There are longer term effects, such as those described by Dr. Moreno, where it takes seven days before the pellicle will provide the maximum protection against demineralization. My answer to your question is that although some clinical treatments remove pellicle, it does re-form quite quickly and I don't think that its temporary loss would be particularly harmful.

Dr. Bell: I'd like to make one comment and then I have a question. I was pleased to hear that you pre-equilibrate the apatite prior to doing these studies. I think this is particularly important. As many of you are aware, the apatite surface can absorb many species, particularly calcium and phosphorus, and the surface charge will change dramatically depending on pretreatment. It is important, in any adsorption work, that you characterize the surface prior to the study, so that you know what that surface looks like, e.g., you know what charge it has. A question in relation to the affinity constant of statherin on hydroxyapatite: I think it was 11 in the absence of fluoride, and 124 in the presence of 10^{-3}M fluoride and that you interpreted that as the fluoride replacing hydroxyl. In essence you end up with a fluorapatite surface. But, the affinity constant of the molecule on fluorapatite was only 34 compared with 124 for hydroxyapatite in sodium fluoride. Can you give an explanation for that?

Dr. Hay: As I said in the paper, we would not have expected the affinity constant of statherin on HA, in the presence of 10^{-3}M fluoride, to be greater than the value for FA, under the same conditions (Table 3). I have no explanation for the higher value for HA in the presence of fluoride, but perhaps Dr. Moreno might like to comment.

Dr. Moreno: Dr. Bell, I am not sure I followed what you said. Are you referring to the fact that adsorption onto fluorapatite in the presence of fluoride in solution was higher than in the absence of this electrolyte? Yes, this seems to indicate that with the crystals of fluorapatite suspended in an aqueous solution, you do not have a uniformally fluoridated surface but some hydroxyl groups may also be present. The fluoride in solution apparently makes the crystal surfaces more uniform as far as the fluoride is concerned and the adsorption characteristics for the protein are consequently changed.

Dr. Bell: Just to follow that up, was the ionic strength the same, when you added the 10^{-3}M sodium fluoride? Did you, in fact, change the ionic strength?

Dr. Hay: I should have mentioned that the protein adsorption experiments were done in 0.05M sodium chloride which was required to keep statherin in solution. We did not compensate for the addition of the fluoride, but I think that you will agree that this small change in ionic strength would have little effect.

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Dr. Gray: A question on the mechanism by which statherin might be preventing precipitation of calcium phosphate: Do you think statherin acts by complexing or reacting with the calcium and the phosphate or do you think it might be inhibiting crystal growth analogous to the diphosphonates?

Dr. Hay: First of all, statherin, at a concentration of 1μM, inhibits precipitation from solutions having calcium and phosphate concentrations of 4mM and 10mM respectively. Complexing of calcium or phosphate, with a consequent reduction of ion activities, as a mechanism of action, is therefore hardly possible, since enormous amounts of either of the ions would have to be bound by the proteins to significantly reduce activities. Our data suggest that the mechanism by which statherin acts is rather similar to that proposed for other inhibitors of calcium phosphate precipitation, such as pyrophosphate and diphosphonates. Here, binding of the inhibitors to nuclei or to seed crystal surfaces is considered to block growth sites and so delay precipitation. Our recent work has shown that the molecular segment of statherin, which is responsible for inhibitory activity in seeded systems, is the amino-terminal hexapeptide, which contains two vicinal phosphoserines and other negatively charged residues. Interestingly, although this fragment is more active than the parent molecule in seeded systems, it is very much less active than the parent molecule in systems where precipitation is initiated by homogeneous nucleation. This suggests an important function for the relatively uncharged carboxy-terminal segment of the molecule.

Dr. Ingram: I really have two questions. Did you in fact, determine the Ca/P ratio of the apatite which you prepared? And have you tried to treat the material with very dilute pyrophosphate, perhaps five or six micromolar? I would be interested to know whether this can compete for the potential sites for your statherin.

Dr. Hay: First, the Ca/P ratio was 1.67. Second, we intend to study this kind of competitive adsorption, but we don't have any data yet.

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